9.3 Right turns and load tail swing

Sometimes, when a large load moves to the left at the beginning of a right turn, other vehicles try to take the inside lane, not realizing the load's intent. Furthermore, oversize loads with a larger rear overhang have a tail swing that may be faster as it swings than the forward movement of the load, and may project into oncoming lanes towards the end of the turn. In congested areas, that swing may need traffic control to protect surrounding traffic and roadside infrastructure such as trees and signs. Pilot cars need to anticipate that they may need to protect both the oversize load and the other traffic, and adjust their positions accordingly.

If only one pilot car is used, its positioning in advance of an intersection depends on the load characteristics. Typically the pilot car will be ahead of the load unless the load is long and not very wide, in which case the pilot car will likely be in a rear pilot car position. A two pilot car scenario is shown in the diagrams below.

For a right turn, the rear pilot car starts in the right-most lane, while the load swings wide and begins the turn, to keep other traffic from moving in too close to the corner. Once the oversize load is into the turn to the right, the rear pilot car moves left, to the outer rear corner of the load, to follow the tail-swing through the turn, bringing attention to any encroachment into oncoming traffic lanes.

Figure 9.3A Right turns and load tail swing

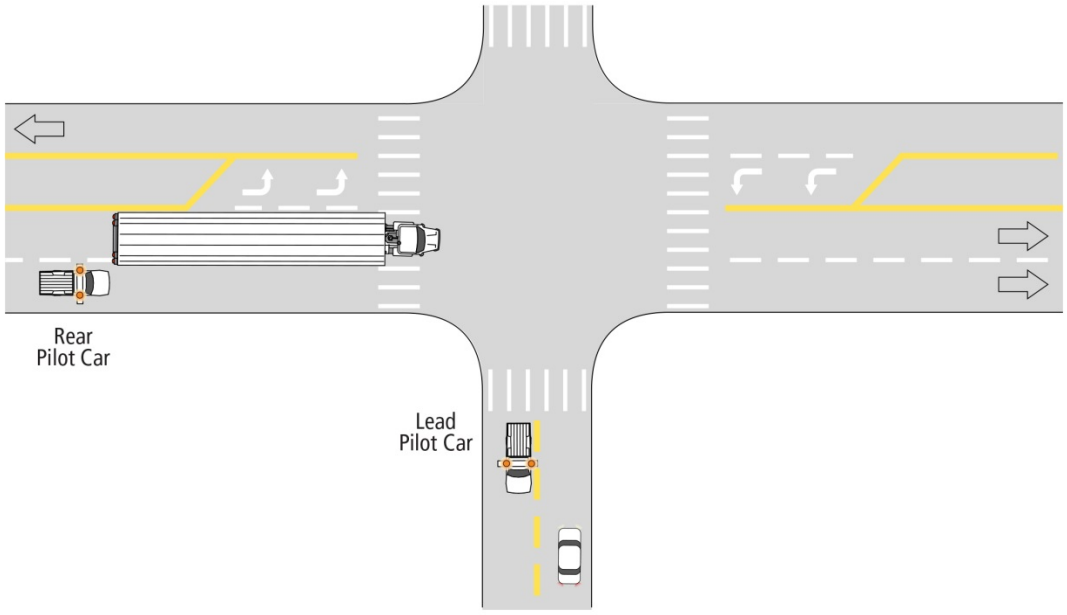


Figure 9.3B Right turns and load tail swing

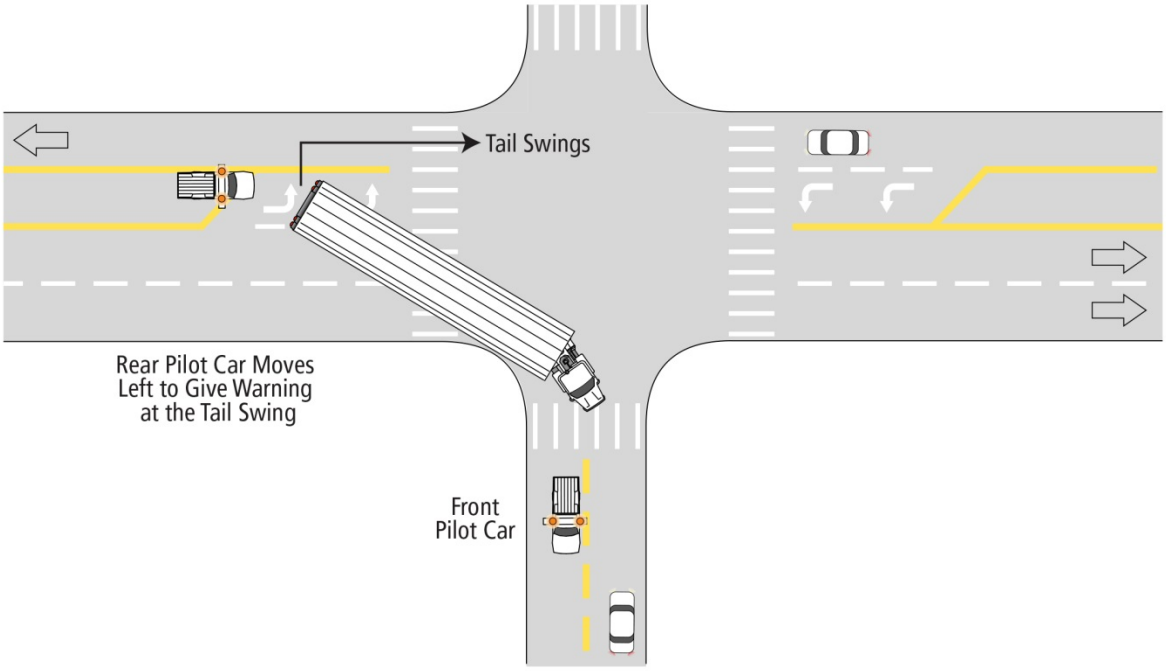
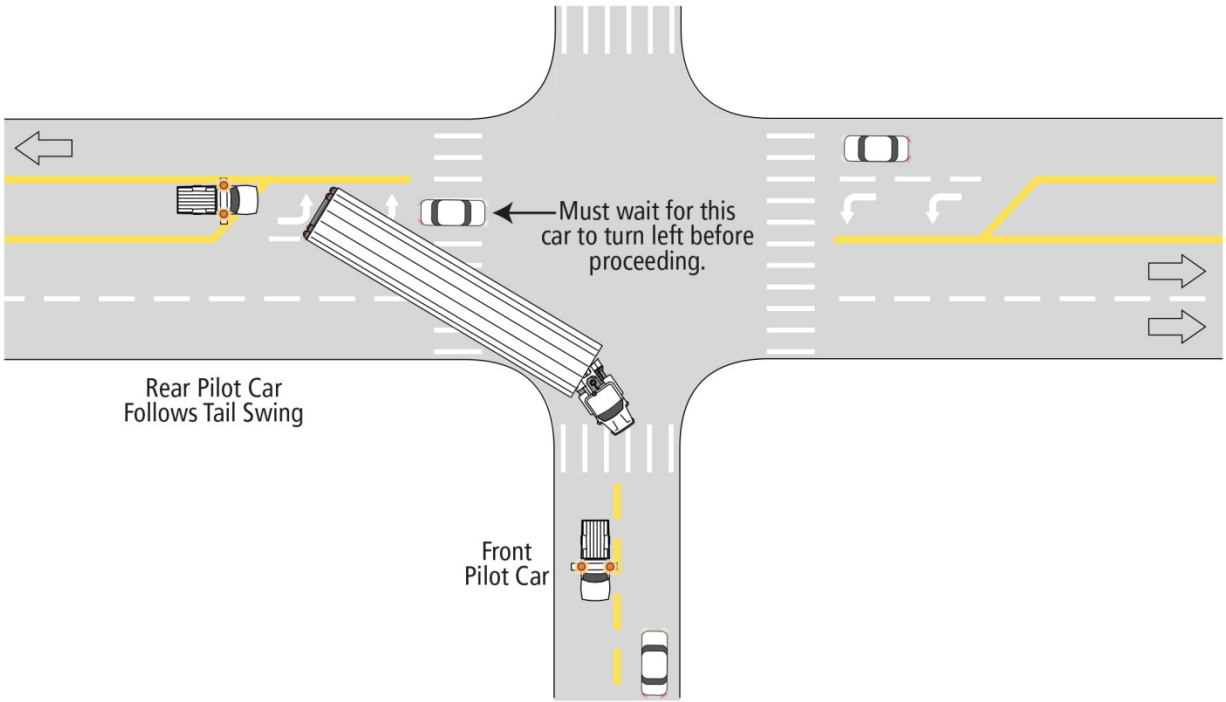


Figure 9.3C Right turns and load tail swing



9.4 Left Turns

See the previous section, 'Right turns and load tail swing', for additional information about manoeuvring the load at intersections.

Figure 9.4A Left turns

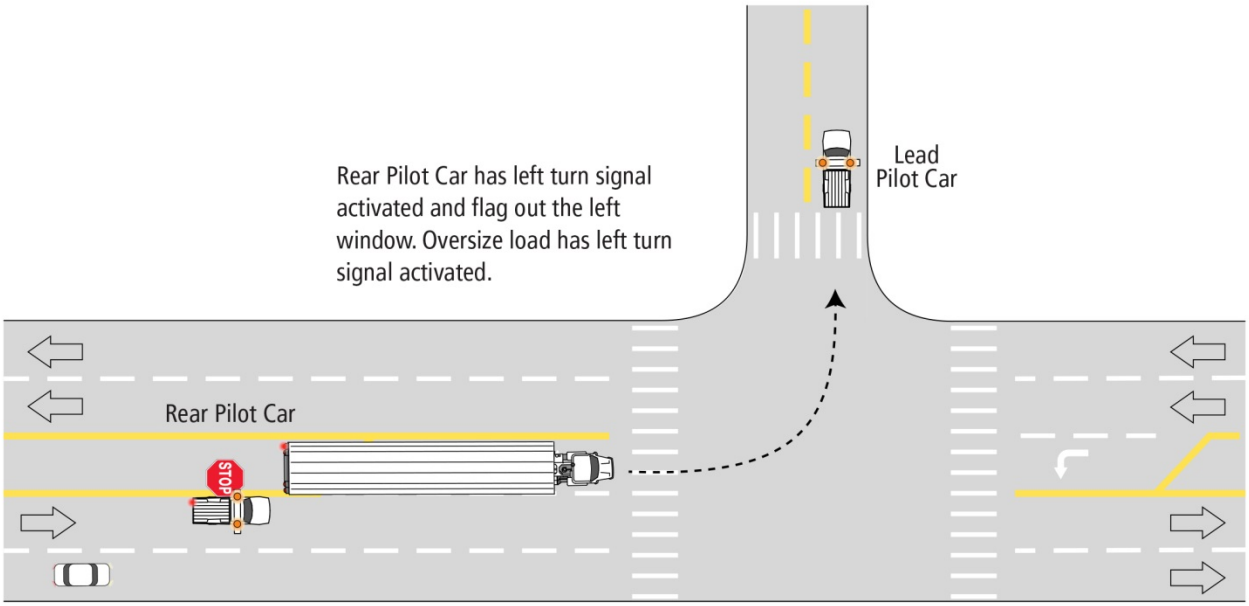


Figure 9.4B Left turns

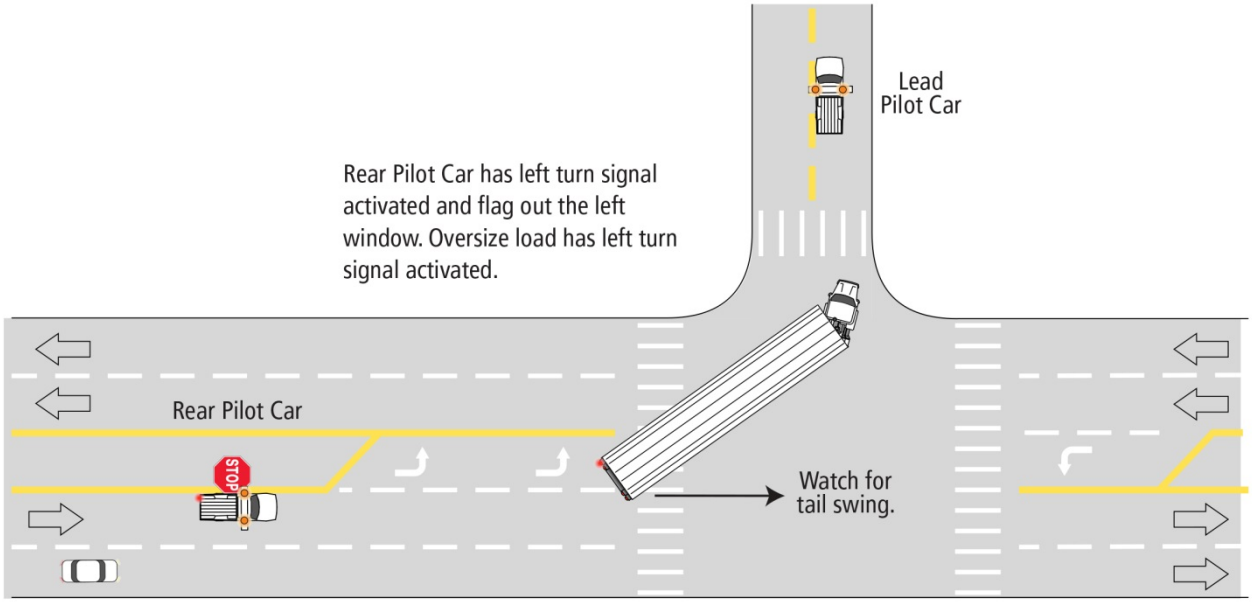
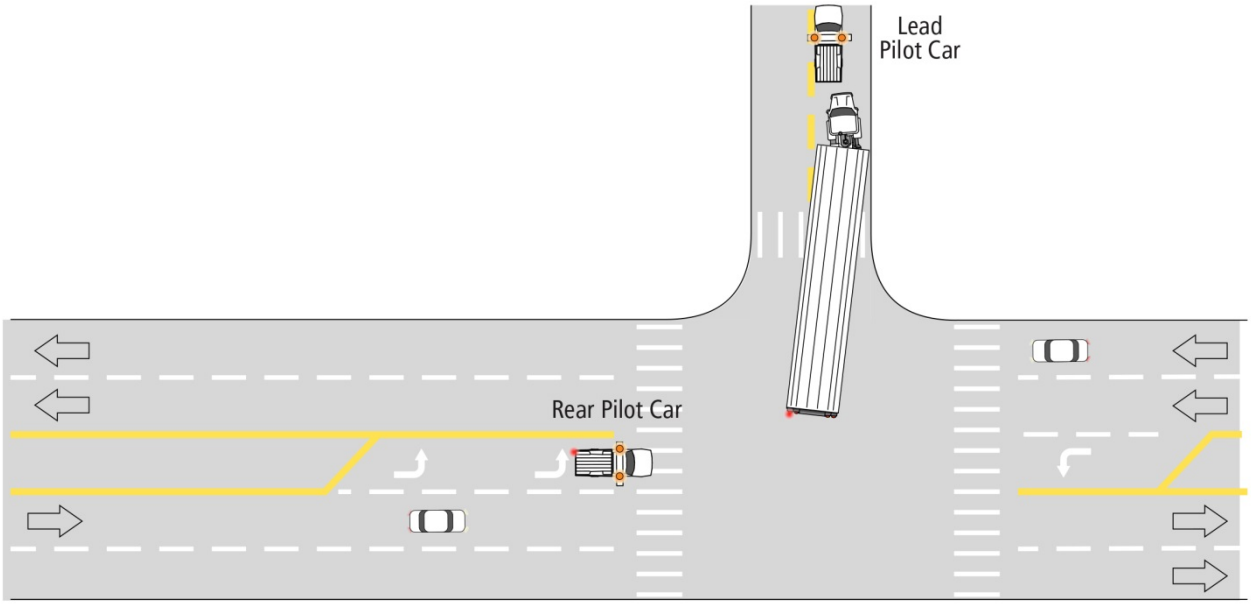


Figure 9.4C Left turns



9.5 Roundabouts

The Ministry has designed roundabouts located on numbered highways to accommodate the same design vehicles as used to design the intersections used on BC highways. When oversize loads are required to traverse through roundabouts, these oversize loads may be required to straddle both lanes of the roundabout and make use of the truck apron, if needed.

If the carrier that will be moving the oversize load on this route has concerns that the load may have difficulty maneuvering through the roundabout, they can provide a schematic of the loaded vehicle configuration to BC MoT's Commercial Vehicle Safety and Enforcement Branch (CVSE) at *commercial.transport@gov.bc.ca*, with a request to check the vehicle's configuration with that of the roundabout's design, to be sure the oversize load will fit. In the unlikely occurrence the oversize load will not be able to maneuver the roundabout, another route will have to be identified.

All BC MoT roundabouts are designed with a truck apron, which is a raised section of pavement around the central island that acts as extra space where a large vehicle trailer can "track." The back wheels of the oversize load's trailer can ride up on the truck apron so the truck can easily complete the turn. This section of the roundabout is specifically for the use of truck trailers and is raised as a means to discourage its use by smaller vehicles.

In British Columbia, most numbered highway roundabouts will have a minimum two lanes (per direction) on the highway legs of the roundabout. Therefore, at these multilane roundabouts, oversize load drivers may occupy the entire circular roadway to travel through the roundabouts. Oversize load drivers should straddle both lanes prior to entering the roundabout, to clearly identify the oversize load will need all lanes to traverse the roundabout. A single lane roundabout is shown in the diagrams below.

Figure 9.5A Operating through roundabouts



Figure 9.5B Operating through roundabouts – making use of the truck apron

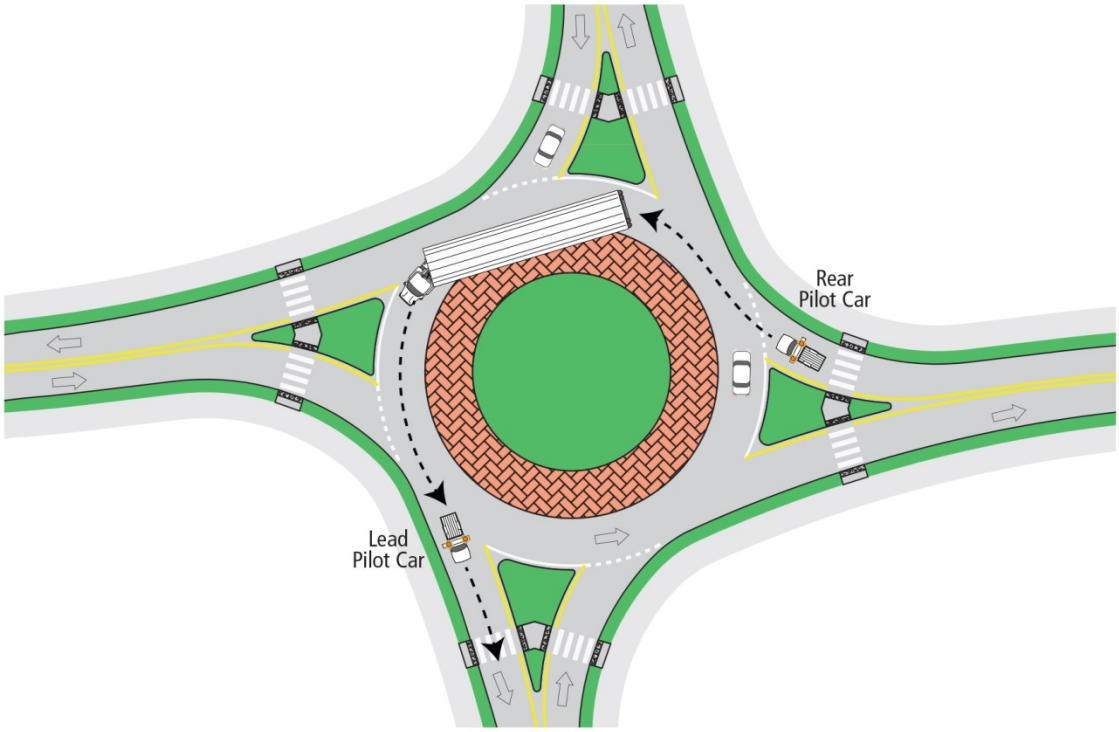
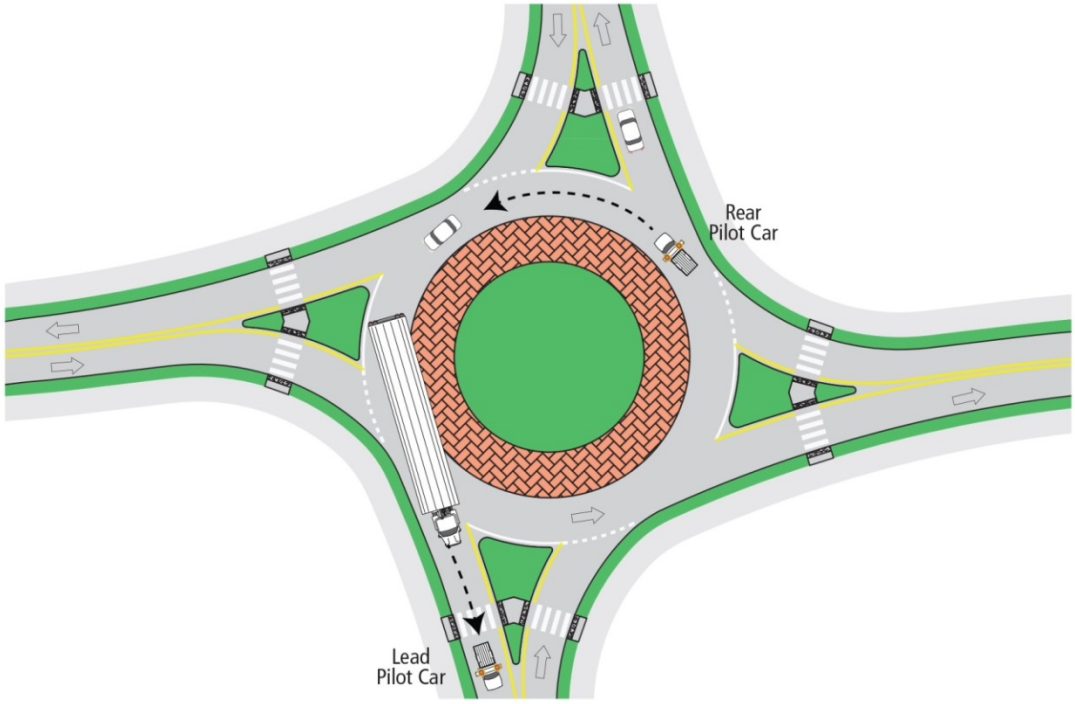


Figure 9.5C Operating through roundabouts



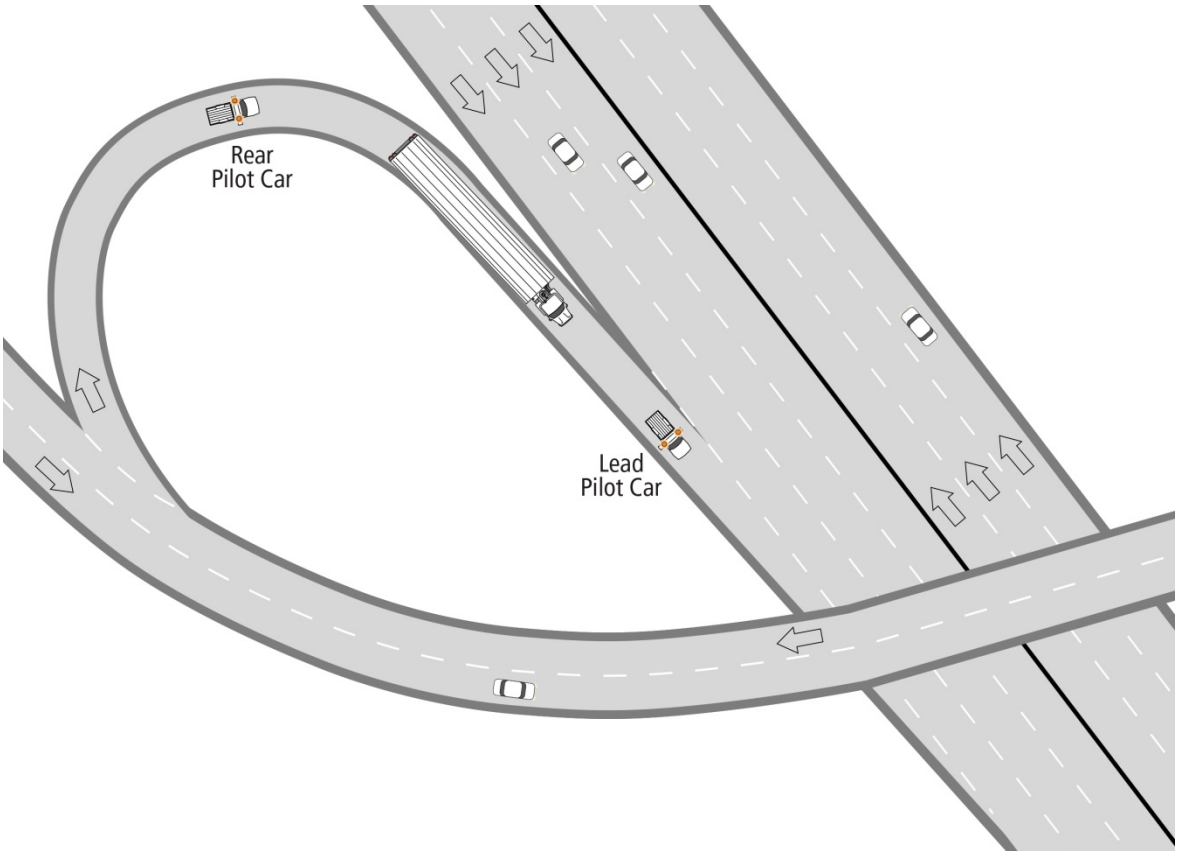
9.6 Interchanges

When approaching an interchange, it is important the move team is familiar enough with the interchange to decide where to position the pilot car(s). In most cases, the pilot car(s) will protect the rear of the load as it moves into the highway it is joining. In some cases, though, it may be useful to have a single pilot car out front to give advance warning of the approach of the load. Good communication within the move team is essential here.

Quite often, cloverleafs are constructed with compound curves due to space limitations, meaning that the tightness and elevation of the curve varies as you move through it. In a large cloverleaf, it is also common to find a zone where one stream of traffic is merging in while another is merging out. For large loads, this means that extra caution about speed and the interaction with other vehicles must be taken.

A two pilot car scenario is shown in the diagram below.

Figure 9.6 Cloverleaf style interchange



Section 10: Pilot Car Load Movement Layouts – Constriction Points

10.1 Overhead Obstacles

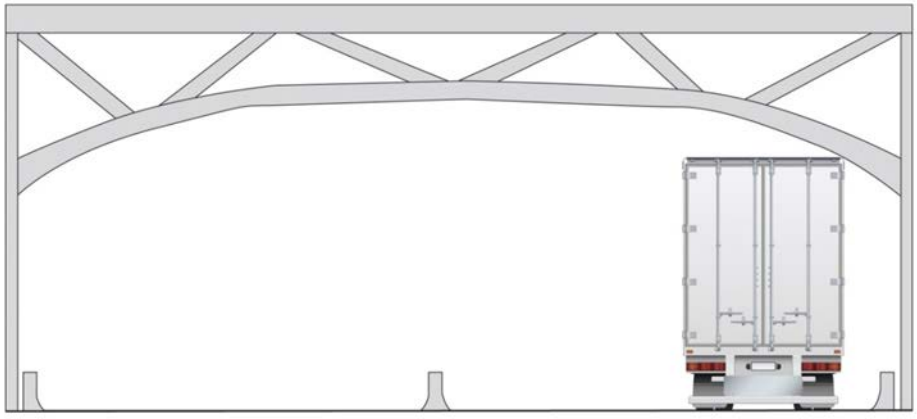
Clearances can vary from one lane to another under a structure. Use caution when approaching an overpass or other type of structure over the roadway, and communicate clearly with the oversize load driver, to ensure that the load you are escorting is in the correct lane.

For loads with heights close to the bridge and other infrastructure clearance restrictions found along the route, checking clearance heights with a laser measure or other device should be considered. When measuring, remember not to make contact with overhead lines. Make any physical measurements to lines to the side of the line so that the measuring stick does not make contact. Checking clearance heights is particularly important at bridges, overpass structures, or power lines that have a variable clearance depending on the lane selected and direction of travel. Other highway infrastructure that may need to be checked along the route for clearances, both vertical and lateral are: signs, luminaire poles, power poles, wires and any device added as part of highway maintenance and construction.

Utility lines can be especially hazardous as many of the lines that cross highway systems are high voltage systems. The handling of telephone, television, or power lines must be left to the professionals from the utility organizations. For the protection of all users of the highway system including those involved with the move, use these precautions:

- Notify utility companies when working near lines when clearances are not known or where clearances need to be adjusted to get the load through
- Treat all wires and electrical equipment that are encountered with the move to be energized. Always check with the utility involved about the status of the system (de-energized or not, etc.)

Figure 10.1 Bridge clearances – Note that available height clearance may vary



10.2 Obstacles on the shoulder

Sometimes shoulder space is limited by manmade structures such as supports for railway overpasses, bridge railings, and signs. It is also common in British Columbia to encounter natural constriction points such as rock outcroppings or other kinds of steep grades. Room on the shoulder may be limited, and when a vehicle, pedestrian, cyclist or other obstruction is on the shoulder, an oversize load may need to move left to avoid the obstacle.

One very important role of a lead or scout pilot car is to communicate about upcoming obstacles to the load driver, so that he or she can adjust speed and positioning as necessary, to get around it. Be as clear and descriptive about the situation to be encountered as possible.

In some cases, it may be necessary to stop the load and do traffic control to make room for the load to move through.

Figure 10.2A Obstacles on the shoulder

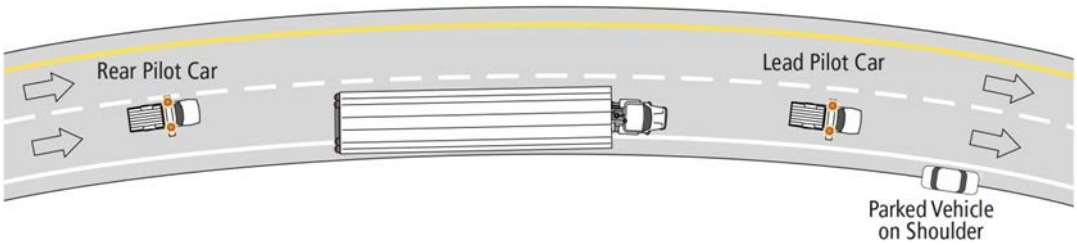


Figure 10.2B Obstacles on the shoulder

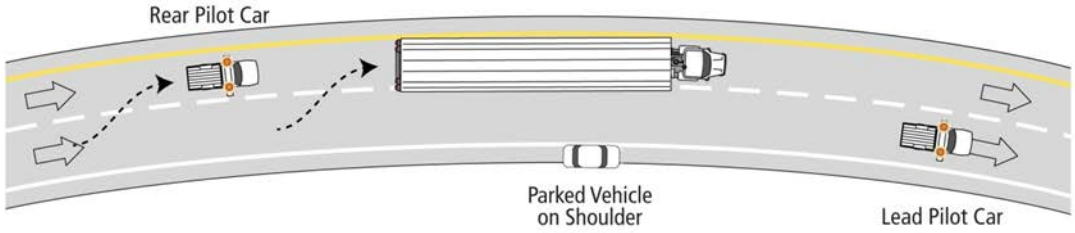
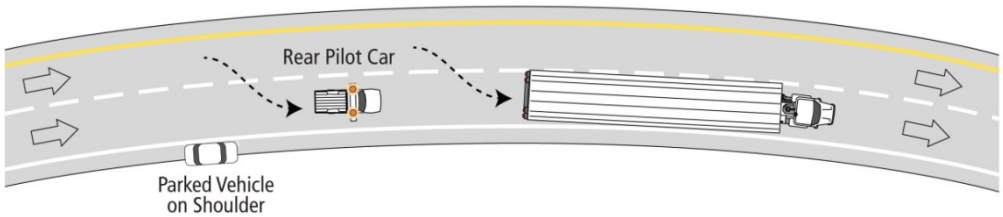


Figure 10.2C Obstacles on the shoulder



10.3 Weigh Scales and Border Crossings

If an open scale or border crossing is encountered during a load move, Pilot Cars should accompany the vehicle they are escorting through the scale or border crossing, and re-enter the highway together, in appropriate positioning.

Figure 10.3A Re-entry from weigh scale

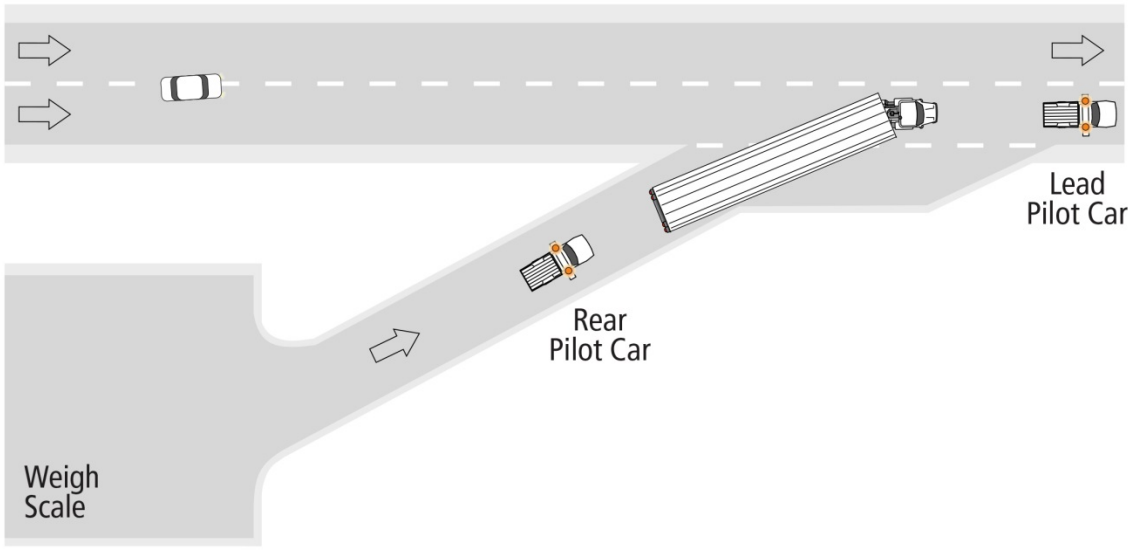
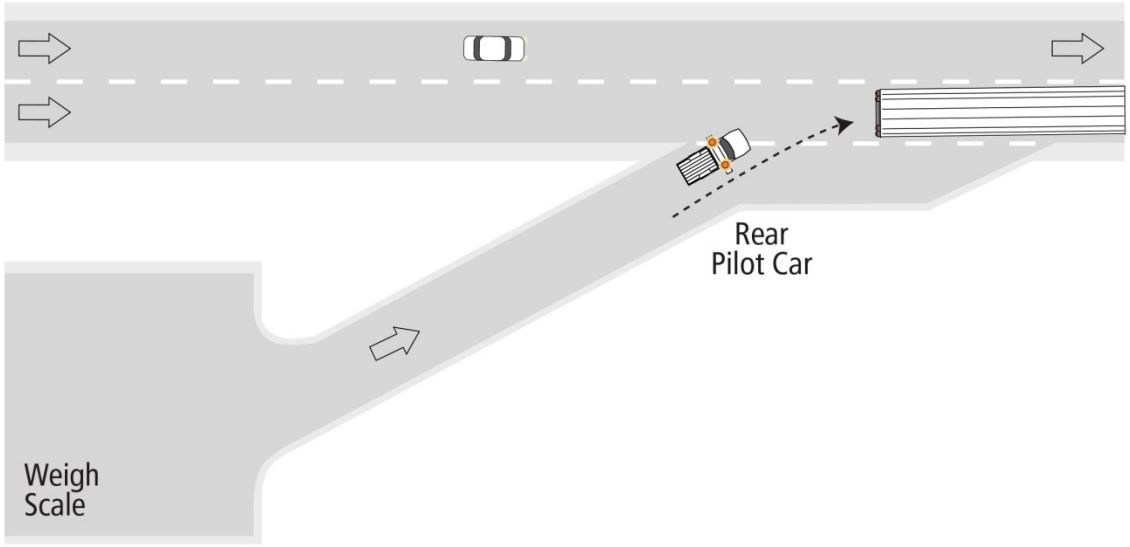


Figure 10.3B Re-entry from weigh scale



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Section 11: Resources

There are many useful links and resources in these guidelines.

11.1 Regulations

- BC Commercial Transport Regulations, Division 7
  http://www.bclaws.ca/civix/document/id/complete/statreg/30_78#division_d2e1638
- BC Commercial Transport Regulations, Division 8.
  http://www.bclaws.ca/civix/document/id/complete/statreg/30_78#division_d2e4480
- WorkSafeBC Occupational Health and Safety Regulation Guidelines Part 18, Section 9a
  https://www.worksafebc.com/en/law-policy/occupational-health-safety/searchable-ohs-regulation/ohs-guidelines/guidelines-part-18
- Part 19 of the Occupational Health and Safety Regulation
  https://www.worksafebc.com/en/law-policy/occupational-health-safety/searchable-ohs-regulation/ohs-guidelines/guidelines-part-19

11.2 Forms

- T-Forms (All)
  http://www.cvse.ca/whatsnew.html
- Extraordinary Load Approval Request Form
  http://www.th.gov.bc.ca/forms/getForm.aspx?formId=1262
- CVSE 1000
  http://www.th.gov.bc.ca/forms/getForm.aspx?formId=1251
- CVSE1011
  http://www.th.gov.bc.ca/forms/getForm.aspx?formId=1258
- CVSE1052
  http://www.th.gov.bc.ca/forms/getForm.aspx?formId=1265

11.3 Manuals and Guidelines

- Commercial Transport Procedures Manual
  https://www2.gov.bc.ca/gov/content/transportation/vehicle-safety-enforcement/information-education/commercial-transport-procedures
- 2015 Interim Traffic Management Manual for Work on Roadways
  https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/traffic-engineering-safety/trafficmanagementmanual/2015trafficmanagementmanual
- Escort Drivers' Handbook (Rev Nov 29, 2011) Alberta Transportation
  http://www.transportation.alberta.ca/Content/docType276/Production/teg054.pdf
- Pilot Car Escort Best Practices Guidelines, SCRA, FHWA, CVSA
  http://ops.fhwa.dot.gov/freight/documents/pilotcar_bpguideline/pilot_car.pdf
- Operation Lifesaver Tips for Professional Drivers
  http://operationlifesaver.ca/resources/professional-drivers/

- ISED-ISDE2366: Mobile Radio Station Licence Application
  http://www.ic.gc.ca/eic/site/smt-gst.nsf/eng/sf06052.html
- Best management practices for mobile 2-way radio use on resource roads in BC, installation and maintenance. BC Ministry of Forest, Lands, and Natural Resource Operations
  https://www2.gov.bc.ca/assets/gov/farming-natural-resources-and-industry/natural-resource-use/resource-roads/bmps_for_radio_use_installation_maintenance_final.pdf
- Important Information on Resource Industry Radio Channels used in British Columbia
  https://www2.gov.bc.ca/gov/content/industry/natural-resource-use/resource-roads/radio-communications

11.4 Websites

- BC Construction Safety Alliance
  https://www.bccsa.ca/Standardized-Traffic-Control-Training.html
- Drive BC
  https://www.drivebc.ca/
- Evergreen Safety Council
  www.esc.org
- CV Height Clearance Tool
  www.drivebc.ca/cvrp/

11.5 Contacts

- Commercial Vehicle Safety and Enforcement Branch (CVSE) at
  commercial.transport@gov.bc.ca
- Provincial Permit Centre at
  onRouteBC.gov.bc.ca *or* 1-800-559-9688

This is the final page.